Antigravity RAG System Architecture Overview

The system uses FAISS as a high-performance vector database.
Chunk embeddings and metadata are saved to disk in the faiss_index directory.
Document provenance is tracked with source file names and 1-based page numbers.

Stage 2 Implementation Details

Document ingestion supports uploading PDFs and chunking text with configurable overlap.
Retrieval fetches the top-k most similar context chunks based on query embeddings.
Answer generation uses evidence grounding to prevent hallucinations.

System Constraints and Fallback Handling

If a user asks about topics not in the uploaded documents, the system states:
I do not have enough information in the provided documents to answer that.
All generated responses include citations in the format: filename (Page X).